



NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Hematopoiesis

PPOM1

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Session Objectives

- 1. Discuss and describe **where hematopoiesis occurs in the fetus, infant, and adult; compare red v. yellow bone marrow; define and evaluate bone marrow cellularity; discuss the monophyletic theory of hematopoiesis; and discuss extramedullary hematopoiesis.**
- 2. Discuss and describe the **concepts of CFU (colony forming units) and the growth factors which stimulate them including the different CSFs (colony stimulating factor), IL (interleukin), erythropoietin, and thrombopoietin.**
- 3. Discuss and describe the **changing morphology and evolution of the myelocyte, monocyte, and lymphocyte beginning with the blast stage and resulting in the mature white blood cell- clinical case examples.**
- 4. Discuss and describe the **changing morphology and evolution of the erythrocyte beginning with the blast stage and resulting in the mature red blood cell – clinical case examples.**
- 5. Discuss and describe the **changing morphology and evolution of the thrombocyte beginning with the blast stage and resulting in the mature platelet – clinical case examples.**

Outline

- Where hematopoiesis occurs
- Normal bone marrow and cellularity
- Extramedullary Hematopoiesis
- **Hierarchy of hematopoiesis**
 - Myelopoiesis
 - Erythropoiesis
 - Thrombopoiesis
 - Lymphopoiesis not discussed in detail, refer to immunology lectures
- Practice questions/case examples

Where does hematopoiesis occur?

– FETUS :

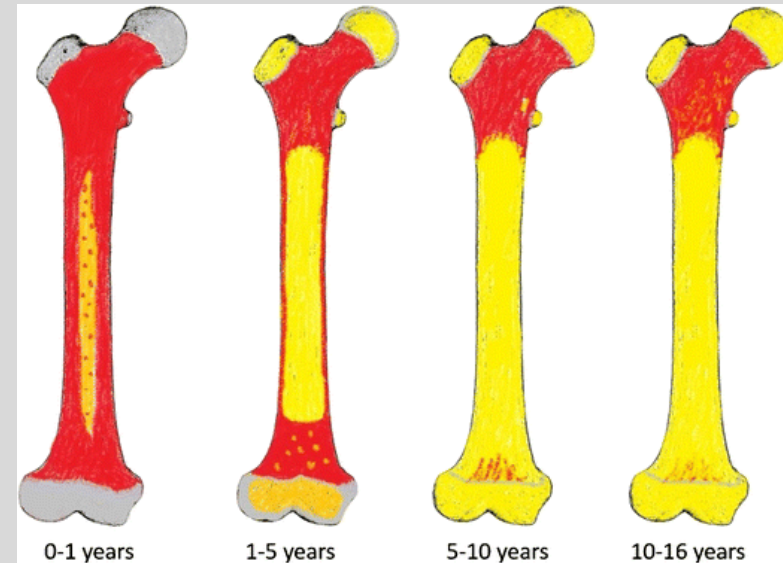
- Yolk Sac – begins in 3rd week of gestation, formation of “blood islands” in wall of yolk sac of embryo.
- Followed by hepatic phase in liver (mostly erythropoiesis)
- Bone Marrow (2nd trimester)

– INFANT: Bone Marrow (all bones)

– ADULT: Bone Marrow (mostly flat bones and proximal femur)

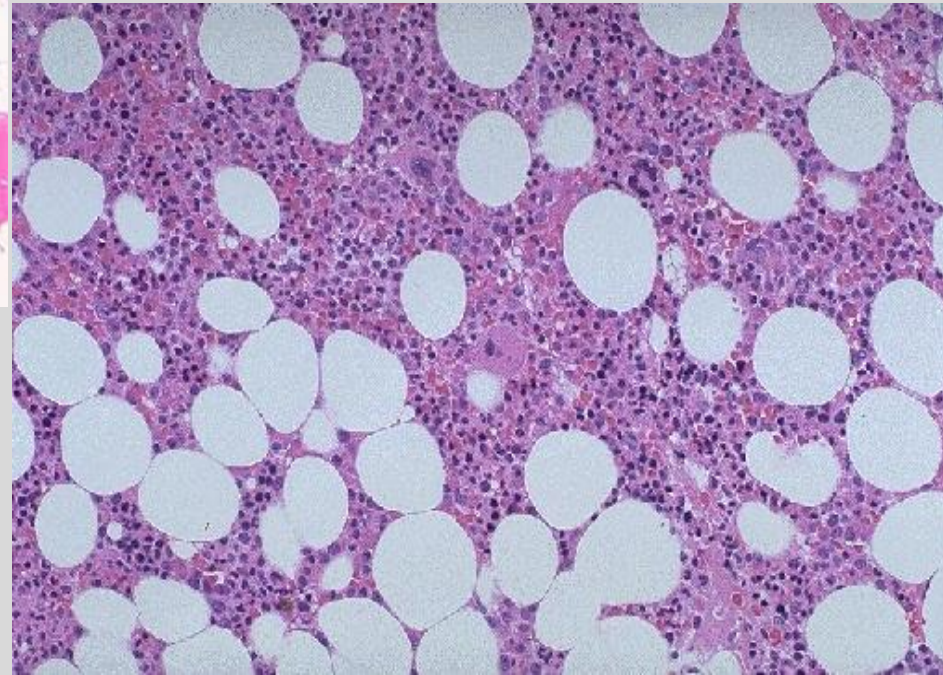
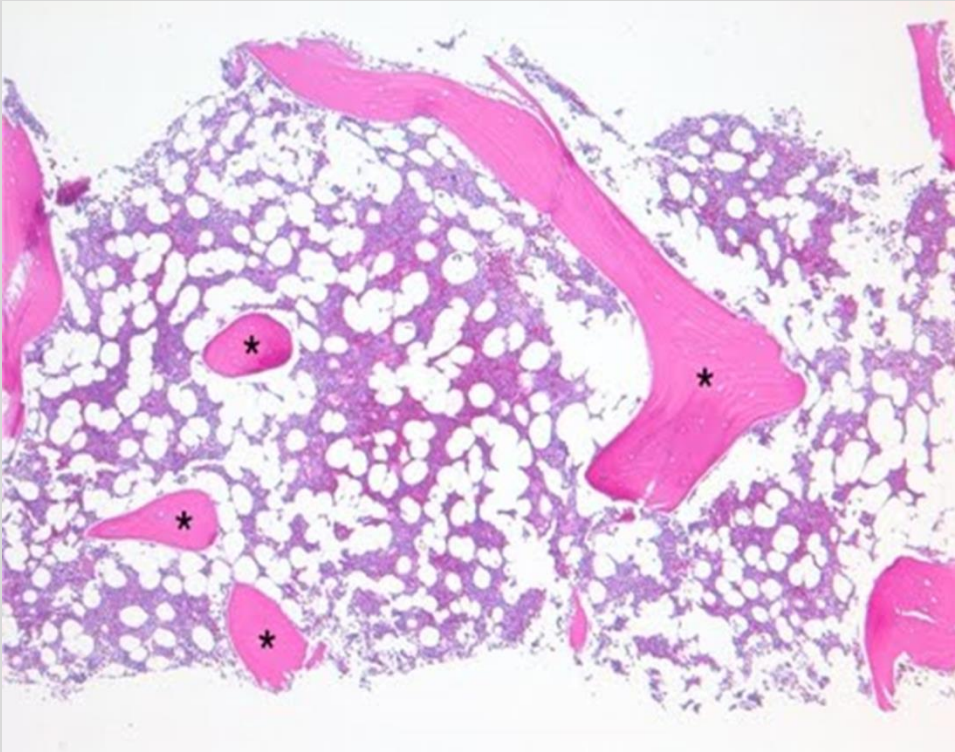
Bone Marrow Histology:

- **Red marrow:**
 - highly vascular
 - composed of stroma, irregular sinusoids, and areas of hematopoietic stem cells
 - site of blood cell formation
 - Also contains stem cells that can produce other tissues in addition to blood cells
- **Yellow marrow:**
 - Filled with adipocytes
 - located in long bones of adults
 - In newborns all the bone marrow is RED and active in blood cell production
 - As child grows, most of marrow changes to yellow

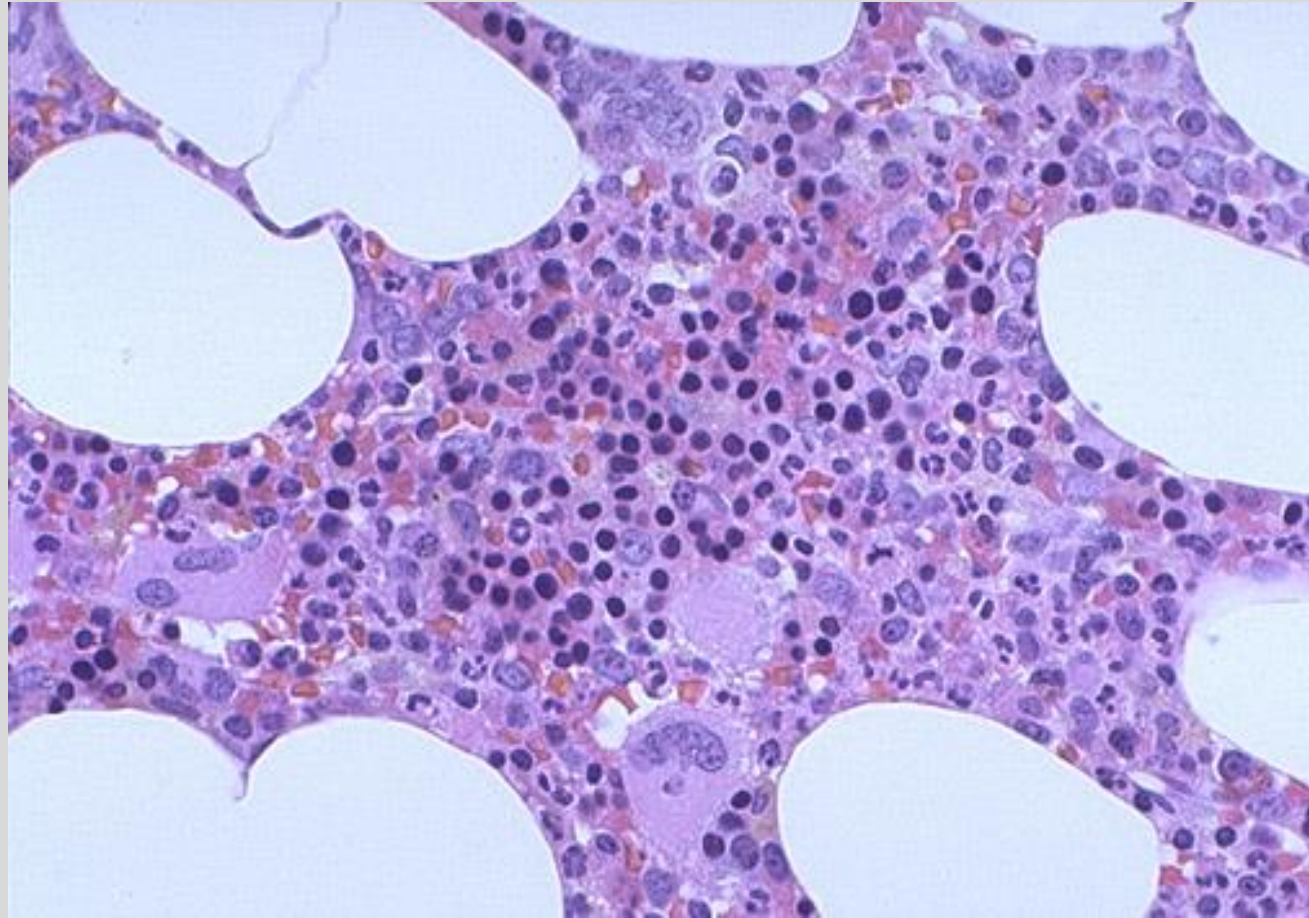


Red Bone Marrow	Yellow Bone Marrow
Red bone marrow consists of RBCs, WBCs, and megakaryocytes in their stages of development.	Yellow bone marrow is named after its appearance as it has a high concentration of fats that appear yellow.
Red bone marrow converts into yellow bone marrow with growth. An adult has less red bone marrow than the yellow one.	Yellow bone marrow rises during adolescents and adulthood and becomes prominent in most bones.
Red bone marrow's other name is medulla ossium Rubra.	Yellow bone marrow's other name is medulla ossium Flava.
The red bone marrow produces crucial cells, including RBCs, WBCs, and Platelets that help us stay alive.	The yellow bone marrow produces fat cells, cartilage, and bones. It stores fat.
Red bone marrow is present at the end of long bones such as shoulder blades, ribs, skull, and spine.	Yellow bone marrow is present in the marrow cavity of most long human bones.

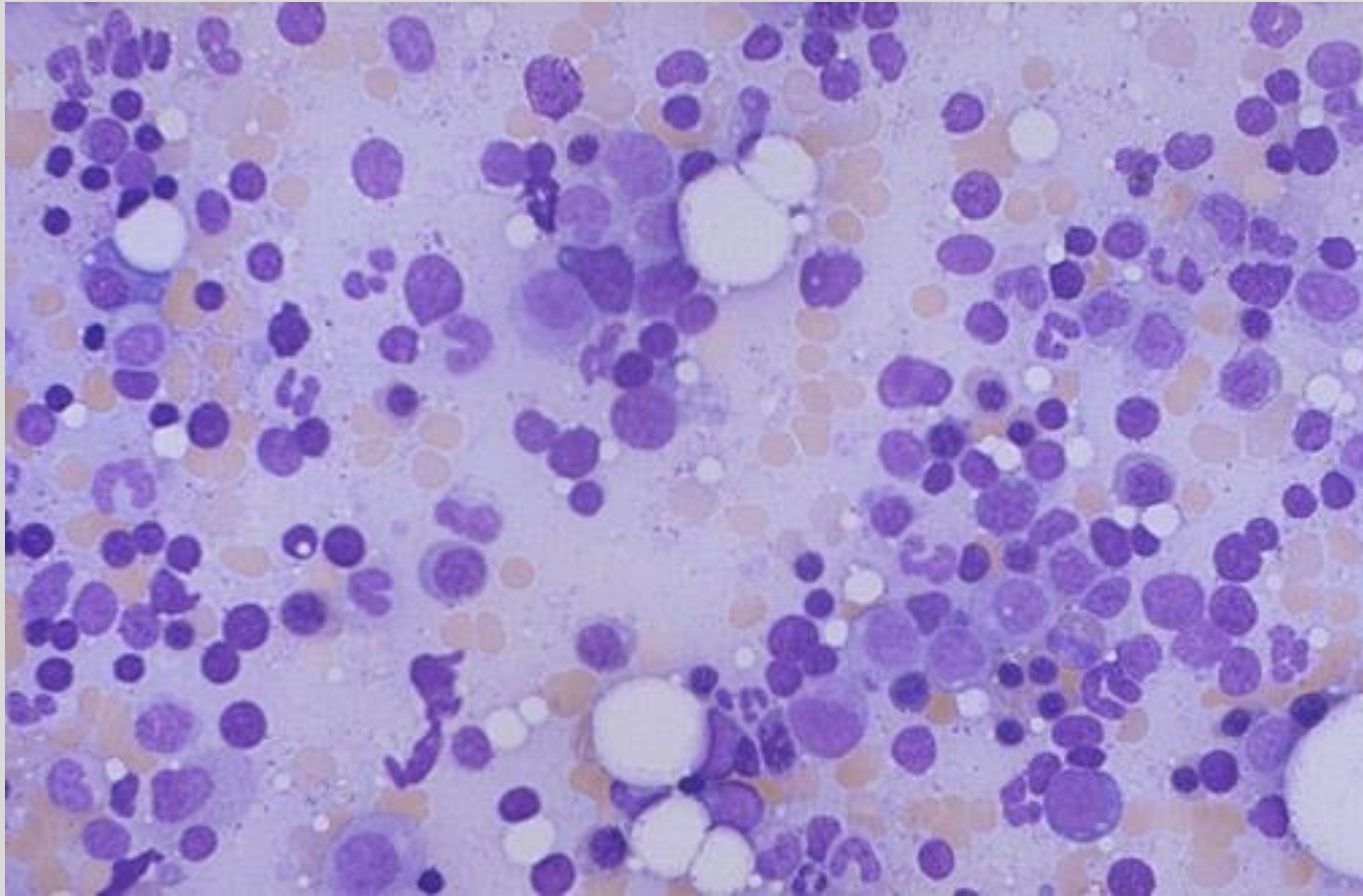
Normal Bone Marrow



Normal Bone Marrow – 3 cell lines (Trilineage)

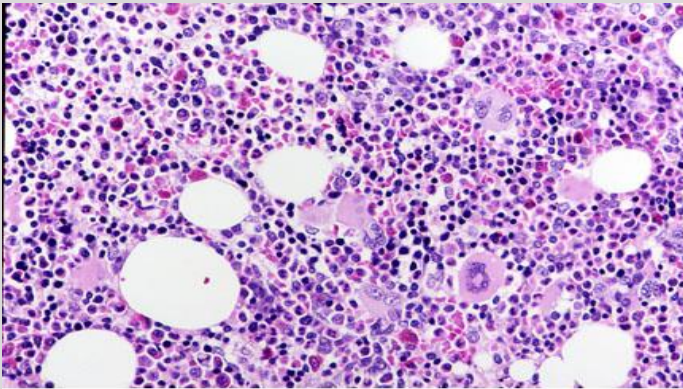


Normal bone marrow smear



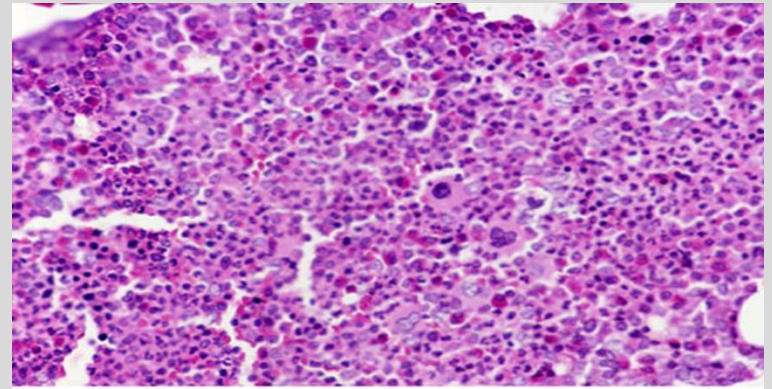
Bone Marrow Histology:

Normocellular:



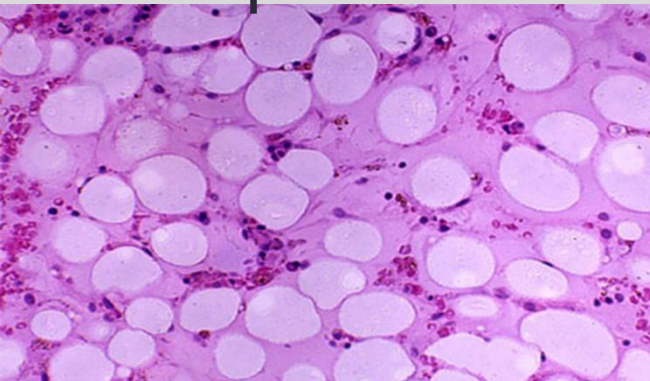
Source: Lichtman MA, Shafer MS, Felgar RE, Wang N:
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Hypercellular



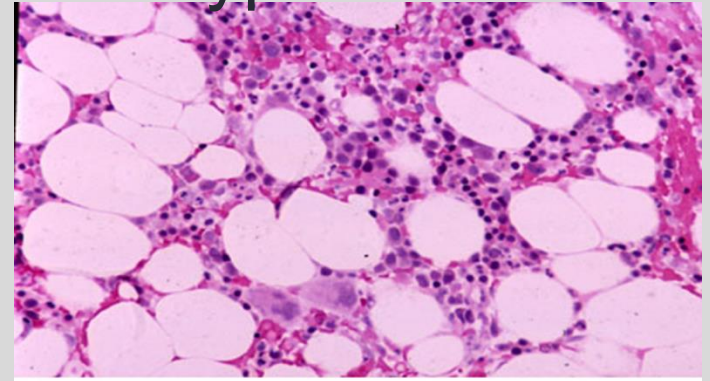
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Aplastic



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Hypocellular



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Normal Bone Marrow

- Consists of blood vessels, sinusoids, and spongelike network of hematopoietic cells
- **Trilineage** maturation (myeloid, erythroid, megakaryocytic lines)
- **Cellularity changes with age:** subtract age from 100 and add +/- 10 %
- $(100 - \text{age}) \pm 10\%$
- Question: 50 yo. male, what's his cellularity supposed to be?
- $(100 - 50) \pm 10\% = (50) \pm 10\% = 40 - 60\%$

Hematopoiesis

- Monophyletic theory of hematopoiesis – all blood cells arise from common stem cell, the **pluripotential stem cell, which gives rise to all other progenitor stem cells**
- Stem cell defined by presence of the CD34+ surface marker protein

Hematopoiesis:

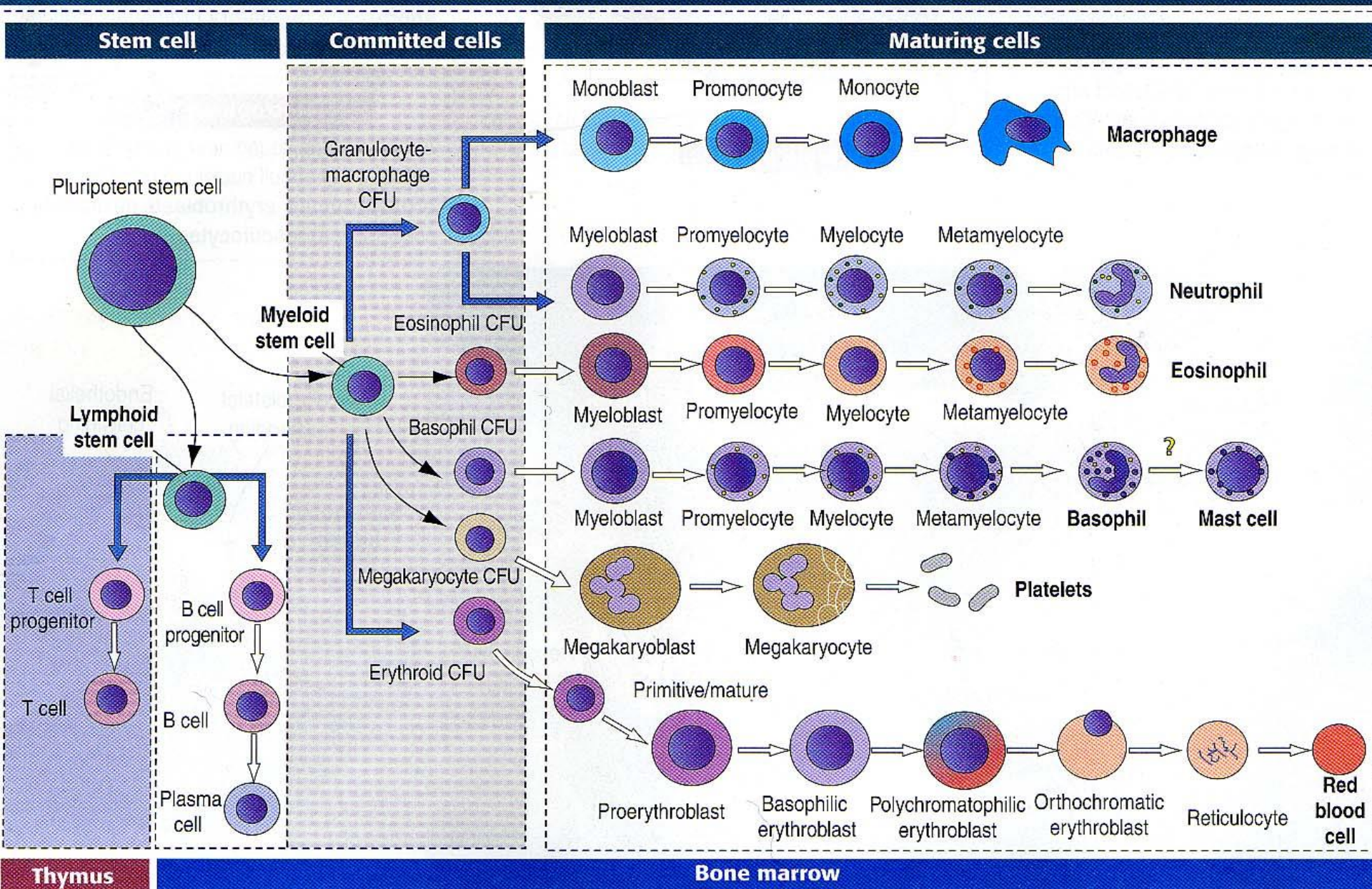
- What if Bone Marrow does not function?
- **Extramedullary hematopoiesis:**
 - Occurs when bone marrow cannot meet the demands for blood cell production
 - **Liver, spleen and lymph nodes** may resume their embryonic hematopoietic activity
 - Observed in **hemoglobinopathies, myeloproliferative disorders, neoplasms involving the bone marrow, and other conditions**

Hematopoiesis: How

- Progenitor stem cells = Colony Forming Units (CFU):
 - Myeloid lineage: (granulocytes, monocytes, erythrocytes, megakaryocytes)
CFU – GEMM differentiates into CFU-GM, CFU-E, CFU-Meg, CFU-Eo, CFU-B
 - Lymphoid lineage: (T-cells, B-cells) **CFU-L**

Lineage	Type of CFU	Peripheral Blood cell produced
Myeloid	Erythroid-CFU (CFU-E)	Erythrocytes
Myeloid	Thrombocytic-CFU (CFU-Meg)	Megakaryocytes
Myeloid	Granulocyte-Monocyte CFU (CFU-GM)	Granulocytes, Monocytes
Myeloid	Granulocyte-Eosinophil CFU (CFU-Eo)	Eosinophils
Myeloid	Granulocyte-Basophil CFU (CFU-B)	Basophils
Lymphoid	Lymphoid lineage CFU (CFU-L)	Lymphocytes

The hematopoietic hierarchy

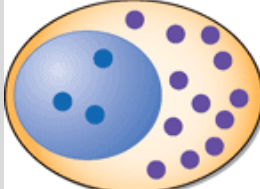
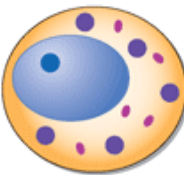
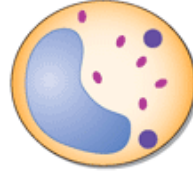
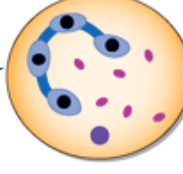


Hematopoiesis: Growth Factors

- **What stimulates the Hematopoietic stem cell to differentiate into the Progenitor cells...and then these to further differentiate?**
- **Hematopoietic growth factors= Colony Stimulating Factors (CSF):**
 - Cytokines stimulate proliferation of progenitor and precursor cells and promote cell differentiation + maturation within specific lineages
 - Cytokine= category of small protein's that are involved in cell signaling
- 3 types of growth factors: (all are cytokines)
 - **Colony stimulating factor (CSF)**
 - **Interleukin**
 - **Erythropoietin + thrombopoietin**

Cytokine	Major activity	Important sources
GM-CSF	Promotes CFU-GM mitosis and differentiation	T cells, endothelial cells
G-CSF	Promotes CFU-G mitosis and differentiation, facilitates neutrophil activity	Macrophages, endothelial cells
M-CSF	Promotes CFU-M mitosis and differentiation	Macrophages, endothelial cells
IL-1	Promotes proliferation of pluripotent hematopoietic stem cells	Macrophages and helper T-cells
IL-2	Stimulates activated T and B cell mitosis, induces differentiation of NK cells	helper T cells
IL-3	Promotes proliferation of pluripotent hematopoietic stem cells	Helper T cells
IL-4	Stimulates B and T cell activation and development of mast cells and basophils	Helper T cells
IL-5	Promotes CFU-Eo mitosis and activates eosinophils	Helper T cells
IL-6	Promotes proliferation of pluripotent hematopoietic stem cells	Macrophages, neutrophils, local endothelial cells
Erythropoietin (EPO)	CFU-E differentiation	Peritubular endothelial cells of kidney capillaries
Thrombopoietin (TPO)	CFU-Meg proliferation and differentiation	Kidney and Liver

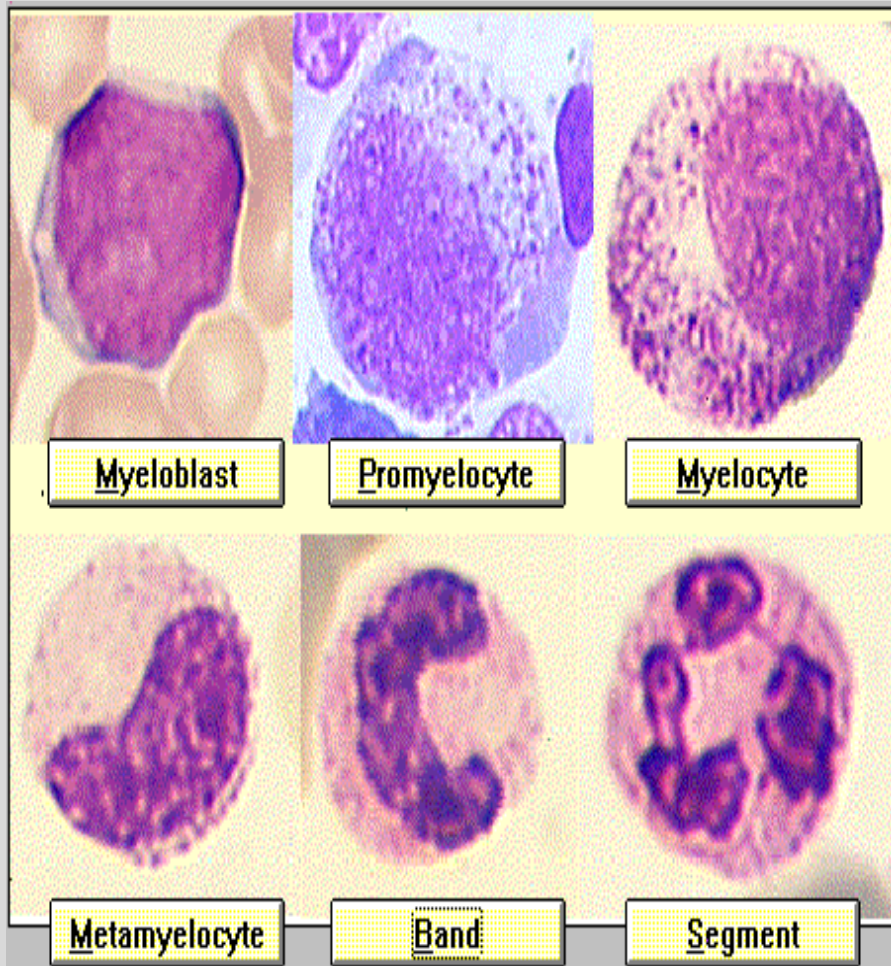
HEMATOPOIESIS: MYELOPOIESIS

	Cell	Stage	Surface Markers ^a	Characteristics
		MYELOBLAST	CD33, CD13, CD15	Prominent nucleoli
		PROMYELOCYTE	CD33, CD13, CD15	Large cell Primary granules appear
		MYELOCYTE	CD33, CD13, CD15, CD14, CD11b	Secondary granules appear
		METAMYELOCYTE	CD33, CD13, CD15, CD14, CD11b	Kidney bean-shaped nucleus
★		BAND FORM	CD33, CD13, CD15, CD14, CD11b CD10, CD16	Condensed, band-shaped nucleus
★		NEUTROPHIL	CD33, CD13, CD15, CD14, CD11b CD10, CD16	Condensed, multilobed nucleus

^aCD= Cluster Determinant; ● Nucleolus; ● Primary granule; ● Secondary granule.

Source: Longo DL, Fauci AS, Kasper DL, Hauser SL, Jameson JL, Loscalzo J: *Harrison's Principles of Internal Medicine*, 18th Edition: www.accessmedicine.com

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MYELOBLAST - Earliest recognizable neutrophil precursor in bone marrow: Large, spherical nucleus. 3 to 5 nucleoli, small amount of **agranular** cytoplasm, stains basophilic (blue)

PROMYELOCYTES – Large spherical nucleus with azurophilic (**primary**) granules, **only made in promyelocytes**

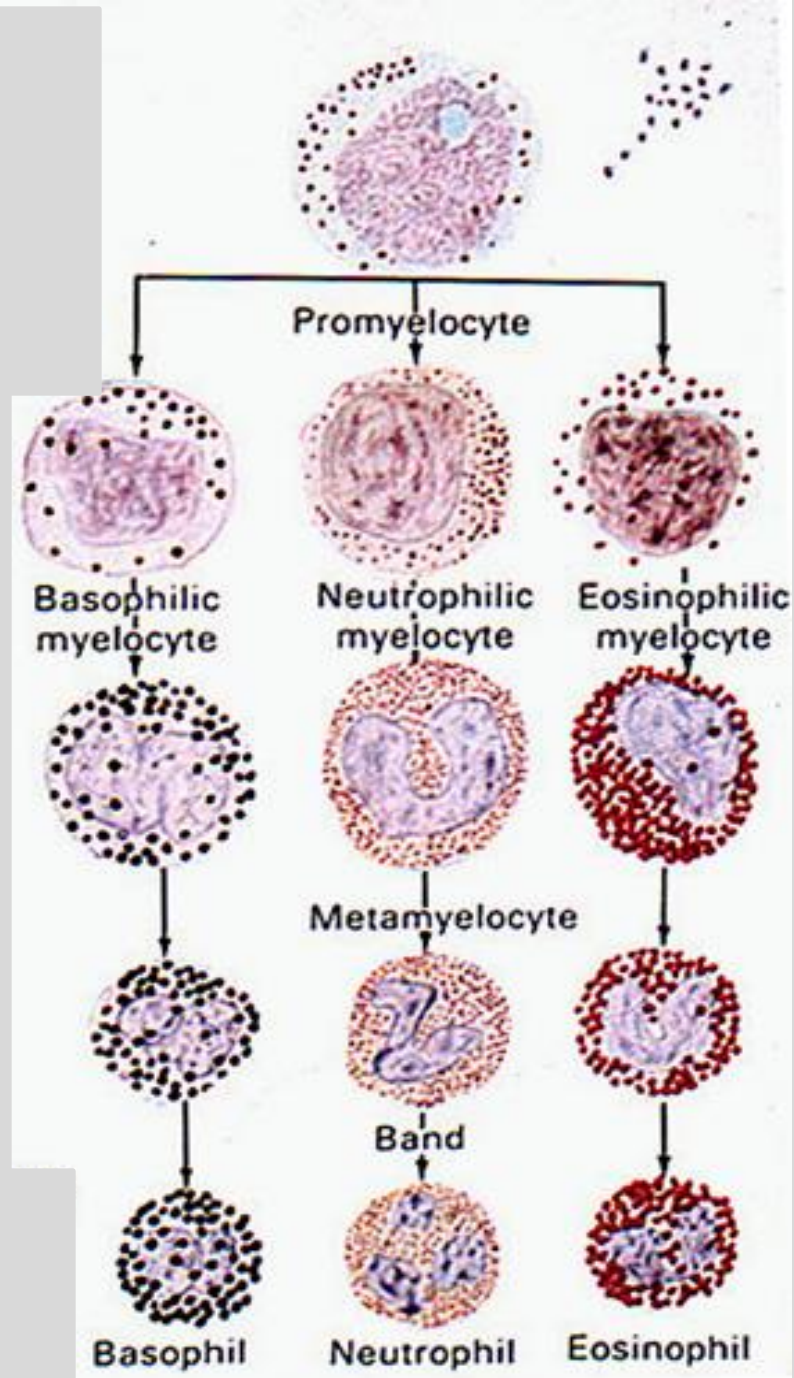
MYELOCYTES - Nucleus becomes more heterochromatic, becomes **indented-Specific granules begin to form**

METAMYELOCYTES – Stage when neutrophil, eosinophil, and basophil can more definitely be identified based on presence of many specific granules- Nucleus indentation deepens becomes kidney-bean shaped

BAND (stab) -Nucleus elongated and horseshoe-like shaped, precedes development of first distinct nuclear lob
-When 2 to 4 nuclear lobes are seen, cell is mature neutrophil or **PMN, i.e., POLYMORPHONUCLEAR NEUTROPHIL**
aka segmented neutrophil

Hematopoiesis: Myelopoiesis

- Myeloblast, promyelocyte, **eosinophilic** myelocyte, eosinophilic metamyelocyte, **eosinophil**
- Myeloblast, promyelocyte, **basophilic** myelocyte, basophilic metamyelocyte, **basophil**
- **Cannot differentiate eosinophilic or basophilic precursors from neutrophilic precursors morphologically until late myelocyte stage or early metamyelocyte stage when specific granules appear**



Hematopoiesis

- Monoblast >>Promonocyte>>Monocyte/Macrophage
- Differentiation and growth of CFU-GM into monocytes is stimulated by GM-CSF, IL-3, and M-CSF.
- Precursors of monocytes in bone marrow are monoblasts and promonocytes
- Monocytes remain in circulation for approximately 16 hours before emigrating to tissues where they differentiate into macrophages

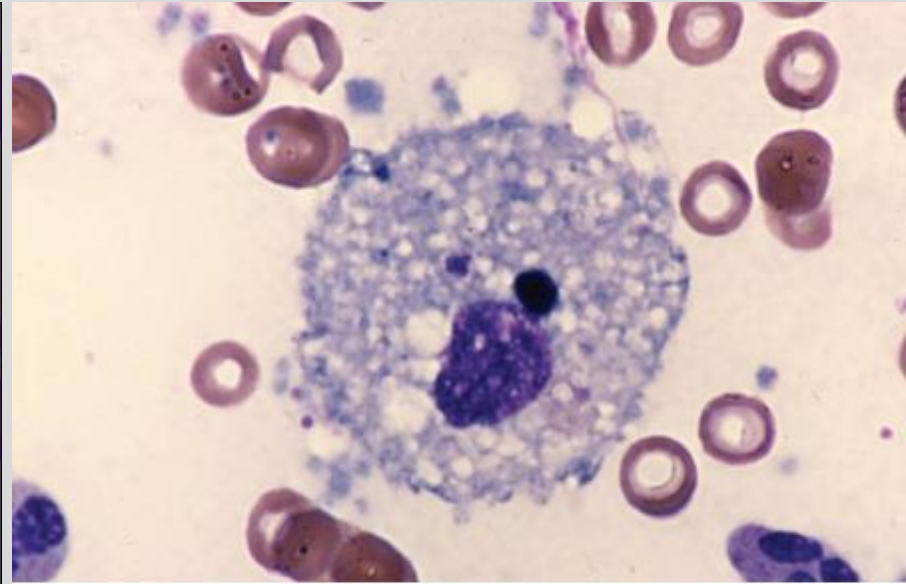
Hematopoiesis:

Myeloid lineage-Agranulocytes

Monocyte + Macrophage



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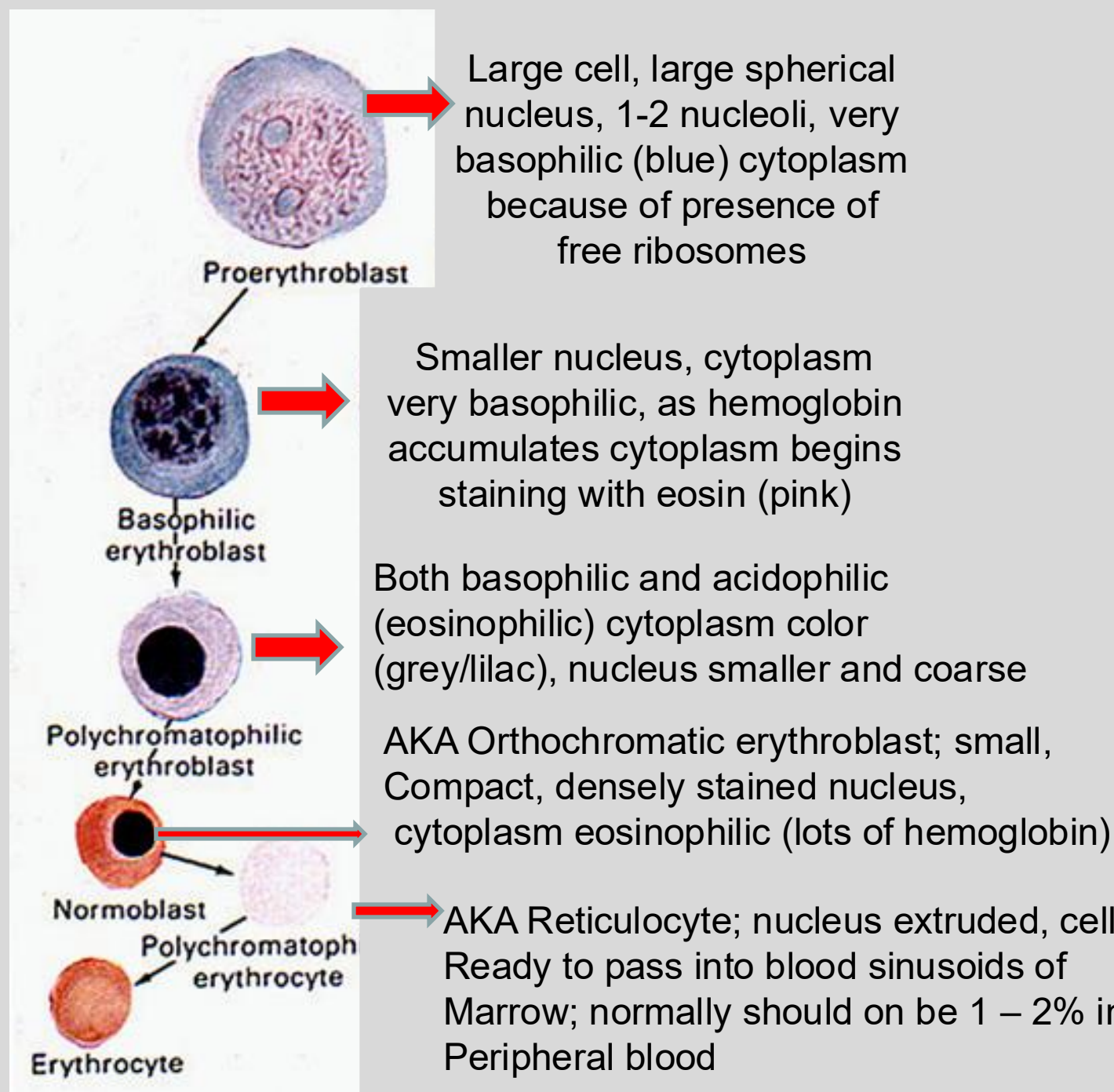
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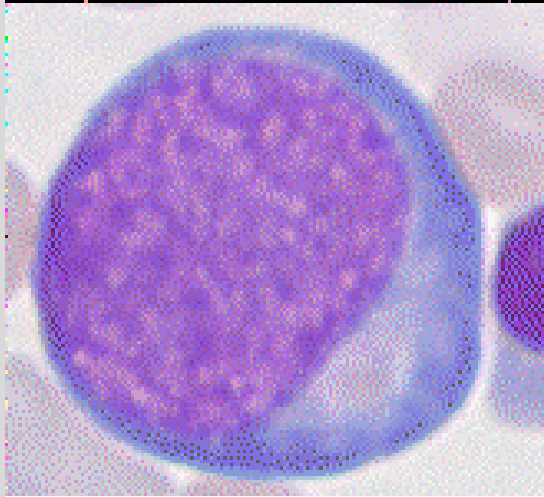
Source: Lichtman MA, Kipps TJ, Seligsohn U, Kaushansky K, Prchal JT:
Williams Hematology, 8th Edition: <http://www.accessmedicine.com>
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Mature monocytes enter the bloodstream, circulate for a few hours, and then enter various tissues where they mature into **Macrophages** (or other phagocytic cells) and function for several months

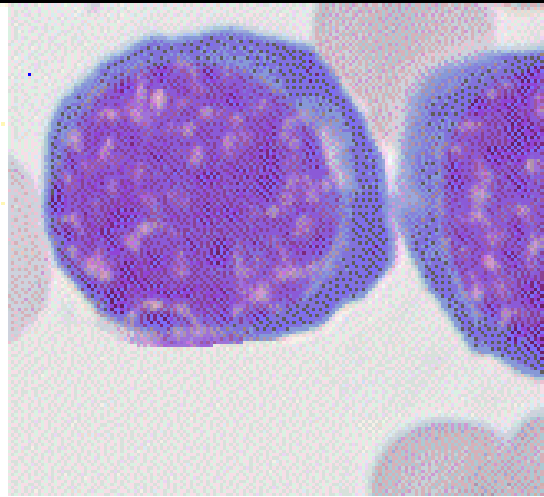
Hematopoiesis – Erythropoiesis

- Under the influence of **erythropoietin**, GM-CSF, IL-3, and IL-4, erythrocytes develop from CFU-GEMM into CFU-E
- Proerythroblast >> basophilic erythroblast >> polychromatophilic erythroblast >> normoblast (orthochromatophilic erythroblast)
>>**reticulocyte** (polychromatophilic erythrocyte)
> erythrocyte (mature red blood cell)

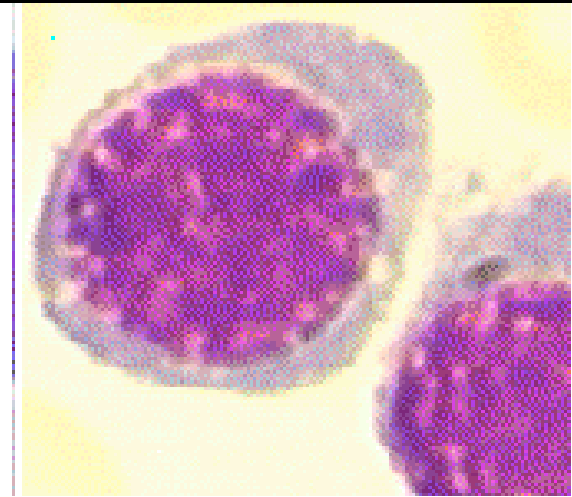




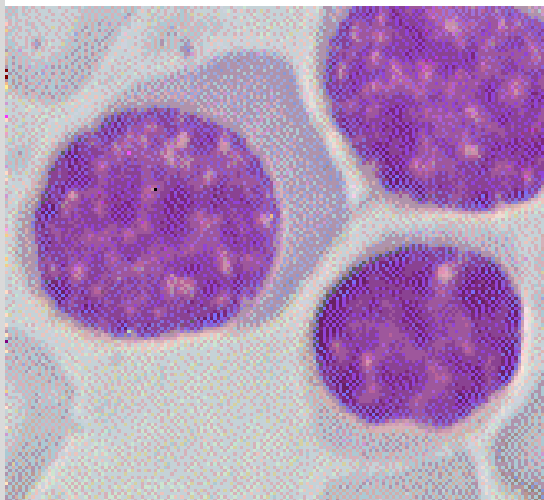
Proerythroblast



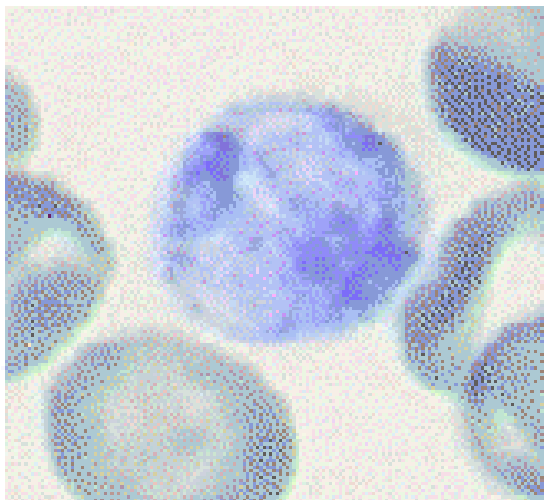
Basophilic Normoblast



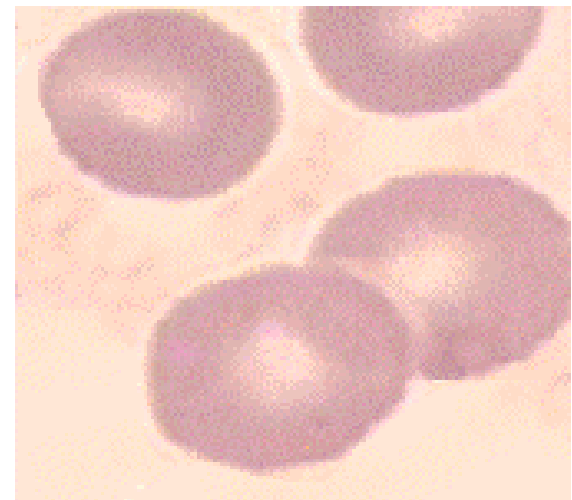
Polychromatic Normoblast



Orthochromatic Normoblast



Reticulocyte



Erythrocyte

AKA Orthochromatic
erythroblast

Hematopoiesis: Thrombopoiesis

Thrombocyte= platelets:

- **Appearance:**
 - Small, anucleate cytoplasmic fragments derived **from megakaryocyte**
- **Action:**
 - Role in Primary hemostatic plug
 - Life-span= 10 days
- CFU-GEMM stem cell differentiates into CFU-Meg. Under the influences of the growth factors:
 - GM-CSF, IL-3, **thrombopoietin**

Megakaryoblast:
Non-lobed nucleus

Pro-megakaryocyte:

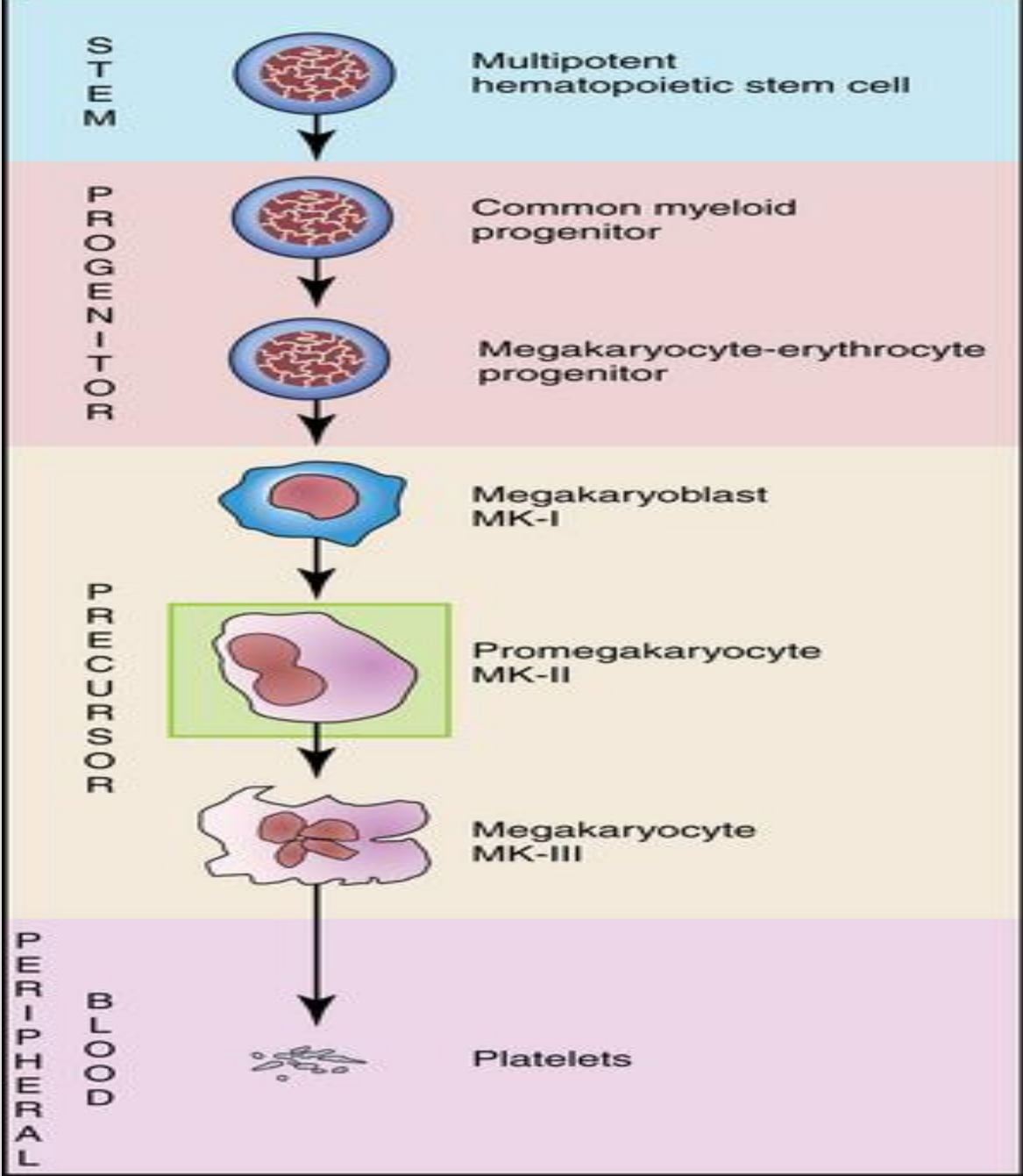
Megakaryocyte
Multilobulated nucleus,
azurophilic granules
Platelets derive from
this stage

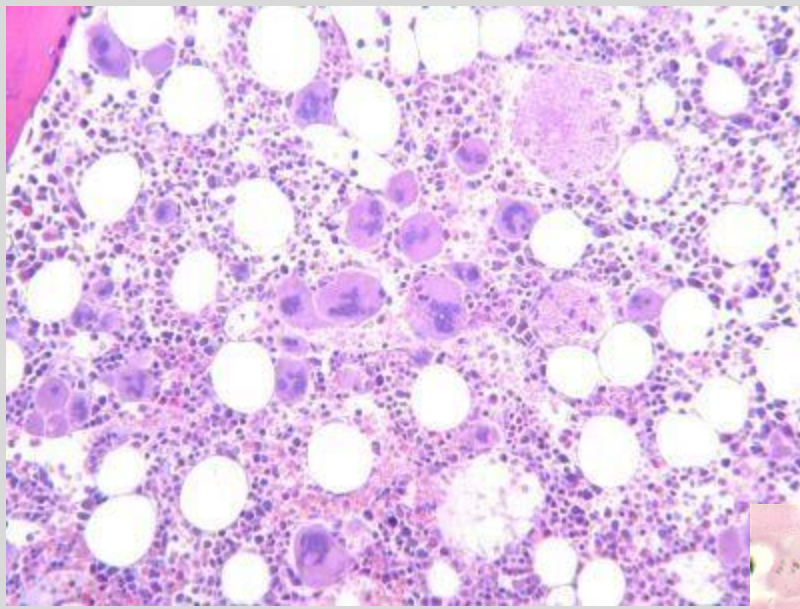
Platelets
Cytoplasmic fragments
of megakaryocyte

Hematopoiesis – Thrombopoiesis

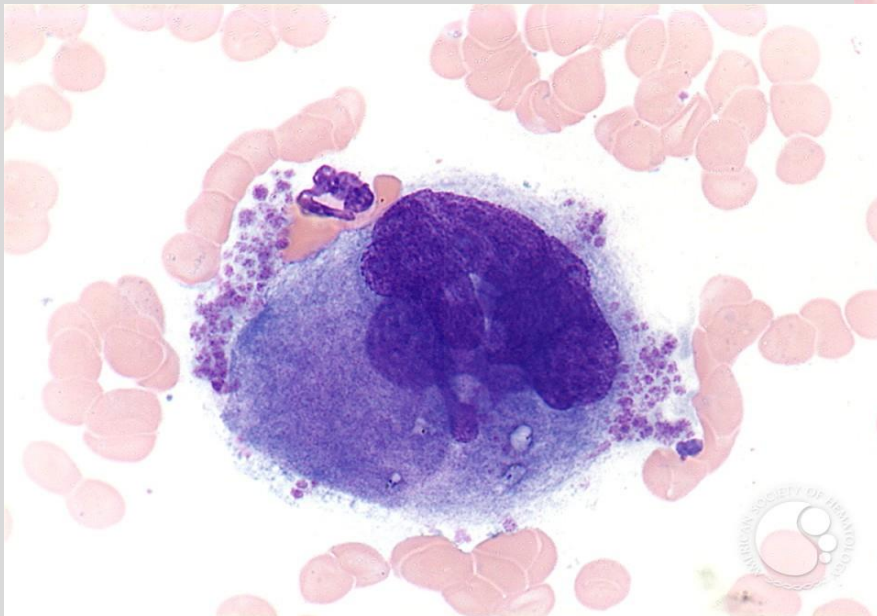
- Platelets produced in marrow from same multipotential stem cells (CFU-GEMM) as erythroid and myeloid series
- Under influence of GM-CSF and IL-3, stem cell differentiates into CFU-Meg, committed to development into megakaryoblast
- **Megakaryoblast large cell with nonlobed nucleus**
- Under stimulation by **thrombopoietin**, ploidy increases before chromosomal replications ceases - cell eventually becomes MEGAKARYOCYTE which then produces platelets (thrombocytes)
- Megakaryocyte has complex, multilobulated nucleus and scattered azurophilic granules. TEM: peripheral cytoplasm megakaryocyte appears divided into small compartments by invagination of plasma membrane – **platelet demarcation channels**

THROMBOPOIESIS





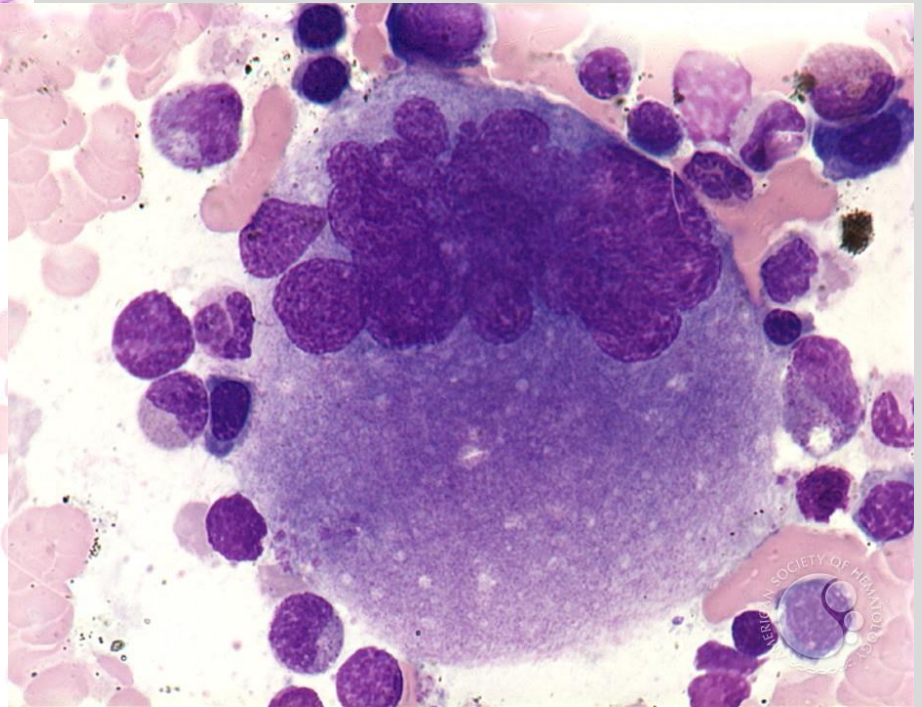
https://atlas.gechem.org/index.php?option=com_k2&view=item&id=1319:loose-clusters-of-megakaryocytes&lang=en



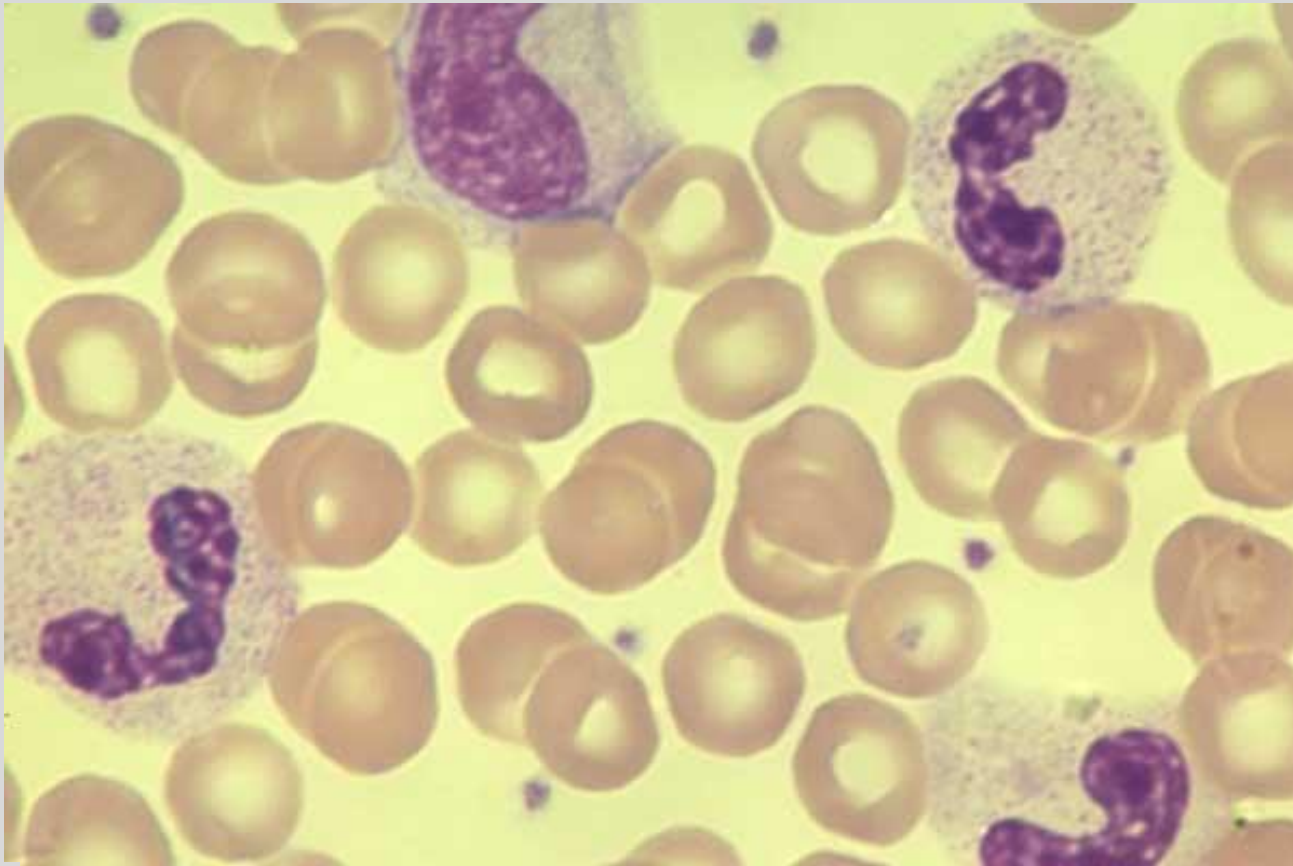
Typical megakaryocyte with platelets budding off the periphery. The granular cytoplasm fragments from separate demarcation regions resulting in the formation of platelets.

<https://imagebank.hematology.org/image/4141/megakaryocyte--1>

Megakaryocytes nuclei have multiple lobes, cytoplasm is granular.



Monocyte vs Neutrophil Band:



Clinical Case

- A 63-year-old man was admitted with a history of gradually increasing abdominal pain of 3 days duration. On admission, the pain was mostly localized to the right upper quadrant (RUQ).
- Labs showed
 - slightly elevated WBC count $12,000/\text{mm}^3$
- The clinical impression of **acute cholecystitis** was confirmed with ultrasonography and computed tomography (CT). A **cholecystectomy** was performed.

Clinical Case

- Which of the white cell lines would most likely be elevated in this **acute** infection of the gallbladder?
 - a) Lymphocytes
 - b) Neutrophils
 - c) Monocytes
 - d) Macrophages
- What does the elevated white blood count represent?
 - a) Infection
 - b) Inflammation
 - c) Stress
 - d) All of the Above

Clinical Case

- A 36-year-old man presented with a 6-day history of bleeding gums and abdominal pain in the left upper quadrant. He reported having had fevers, fatigue, decreased appetite, and unintentional weight loss of 10 lbs during the previous month.
- On physical examination,
 - red, swollen gingivae
 - gingival hyperplasia
 - tender submandibular lymph nodes
 - a palpable spleen
- Labs revealed
 - WBC 194,100 mm³ with 44% blasts
 - Platelets 12,000 mm³

Clinical Case

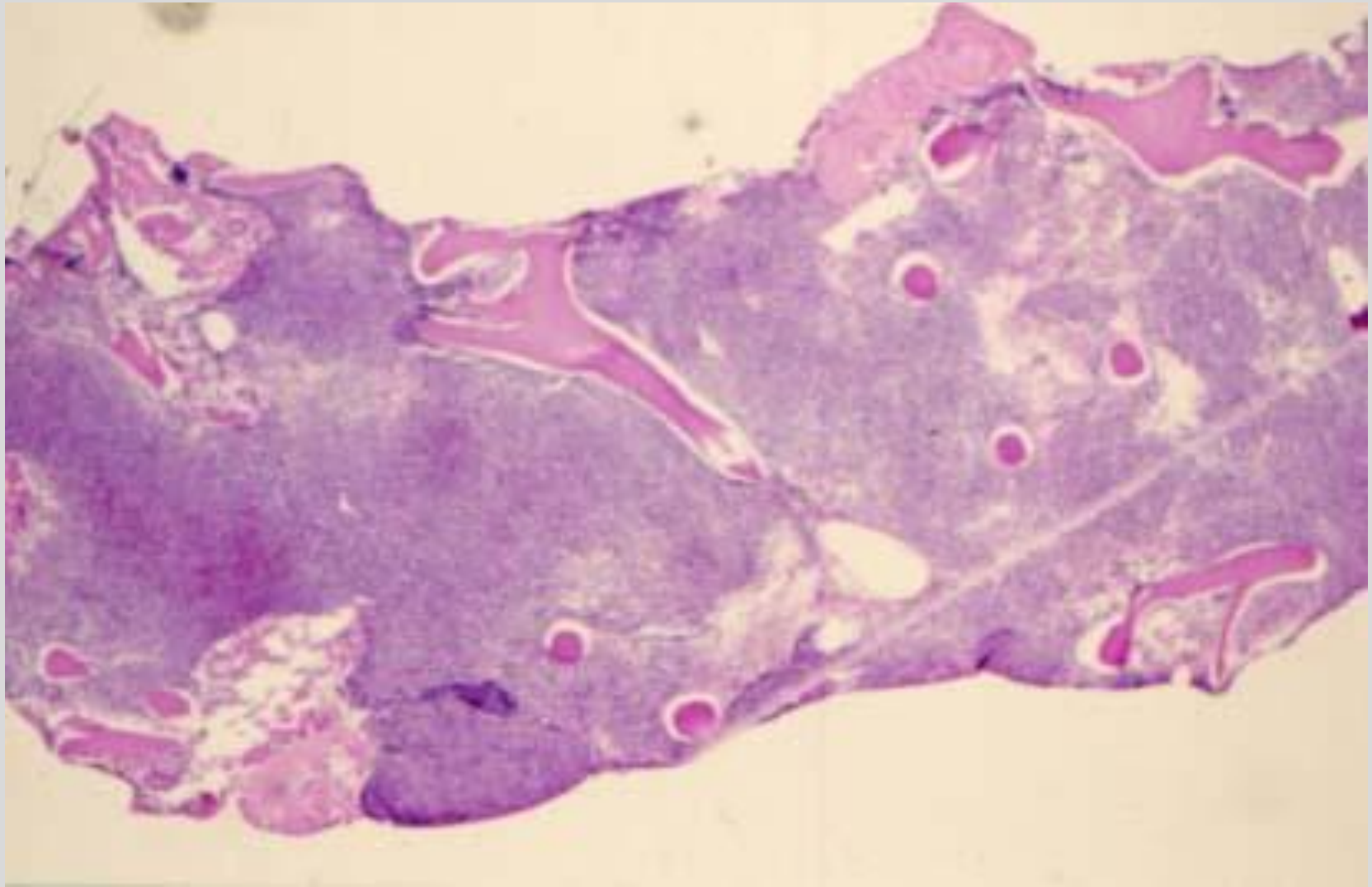


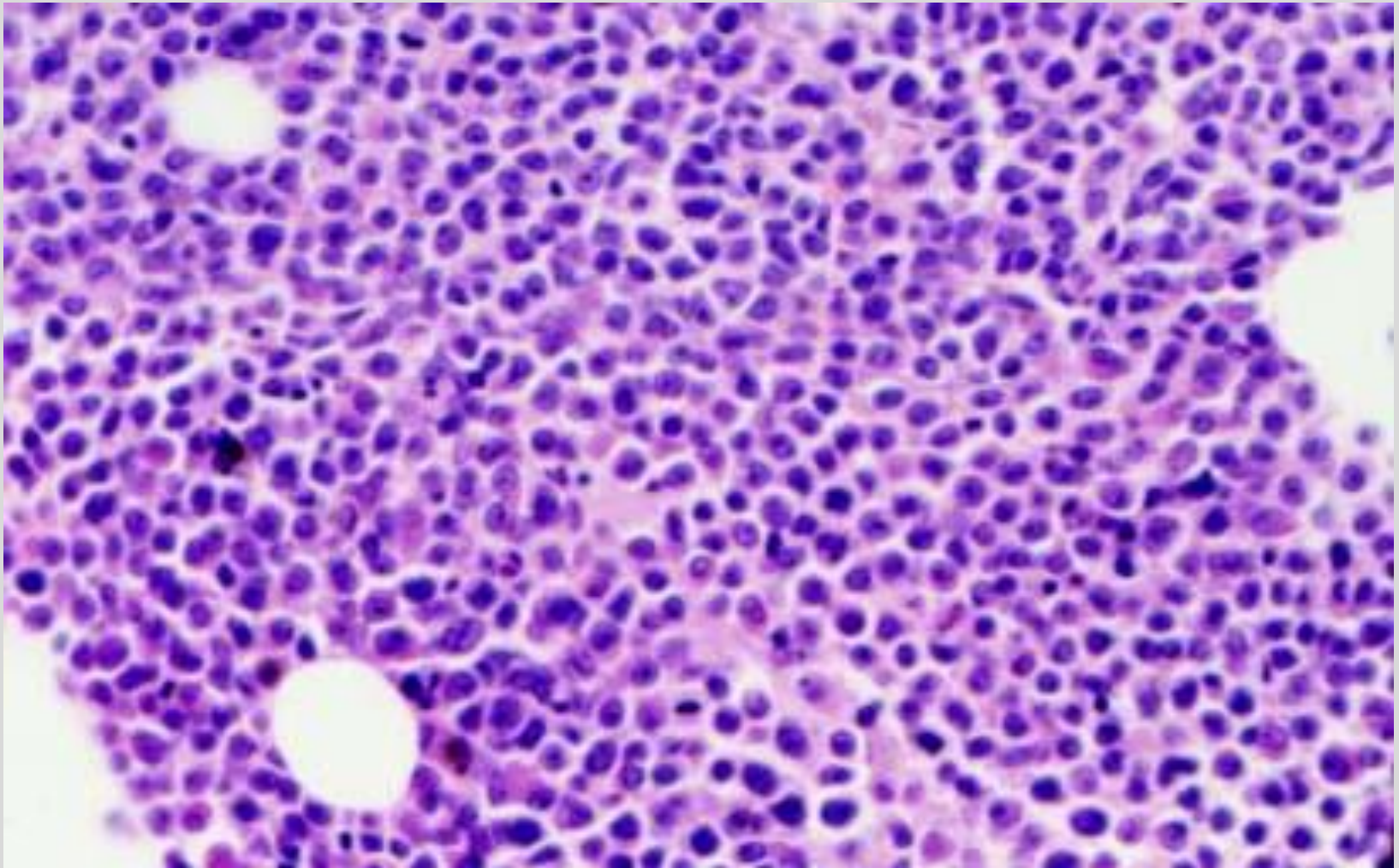
Clinical Case

- Which cell line is responsible for the symptom of *bleeding gums*?
 - a) Monocytes
 - b) Platelets
 - c) B cells
 - d) Plasma Cell
- What hormone/molecule regulates platelet production?
 - a) Erythropoietin
 - b) Thrombopoietin
 - c) IL-1
 - d) IL-3
- With his WBC count (194,000 mm³ with 44% blasts), would you expect that he would be able to fight infections better?

Clinical Case

- A 5 year old white female is brought to her pediatrician because of fever, marked weakness, pallor, Bone Pain, and bleeding from her nose (epistaxis). She has a history of progressively increasing fatigue and recurrent infections.
- Physical Exam shows
 - bruising over the skin
 - sternal tenderness
 - hepatosplenomegaly
 - non-tender lymph node enlargement
- Labs:
 - WBC Lymphocytosis with (30%) blasts,
 - anemia

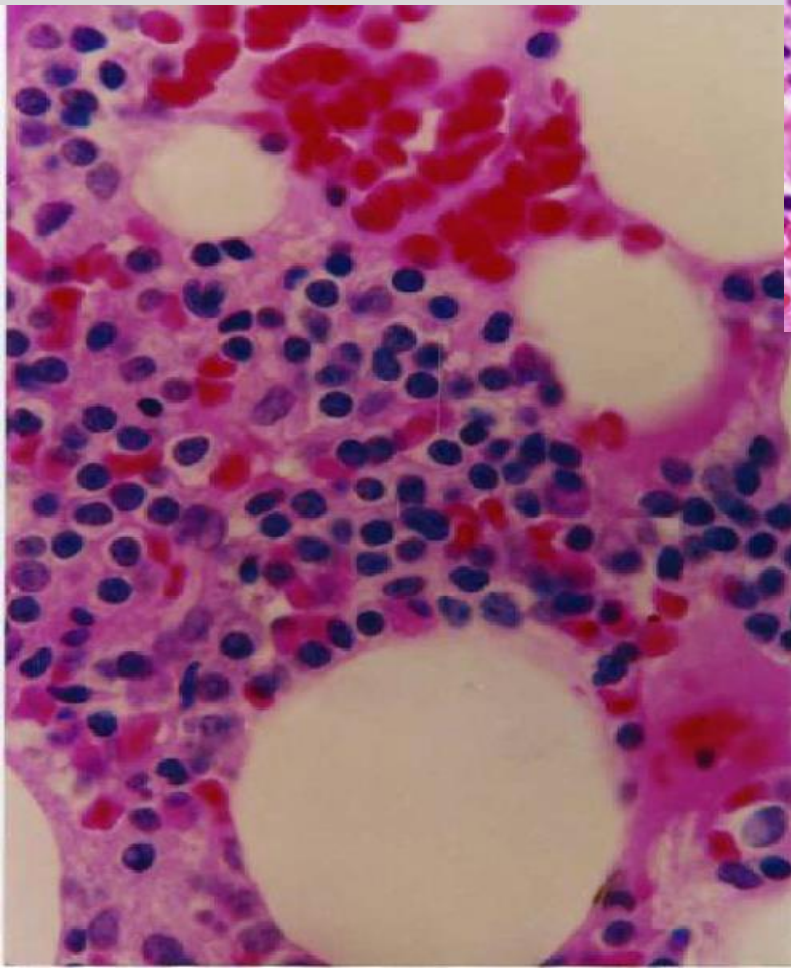
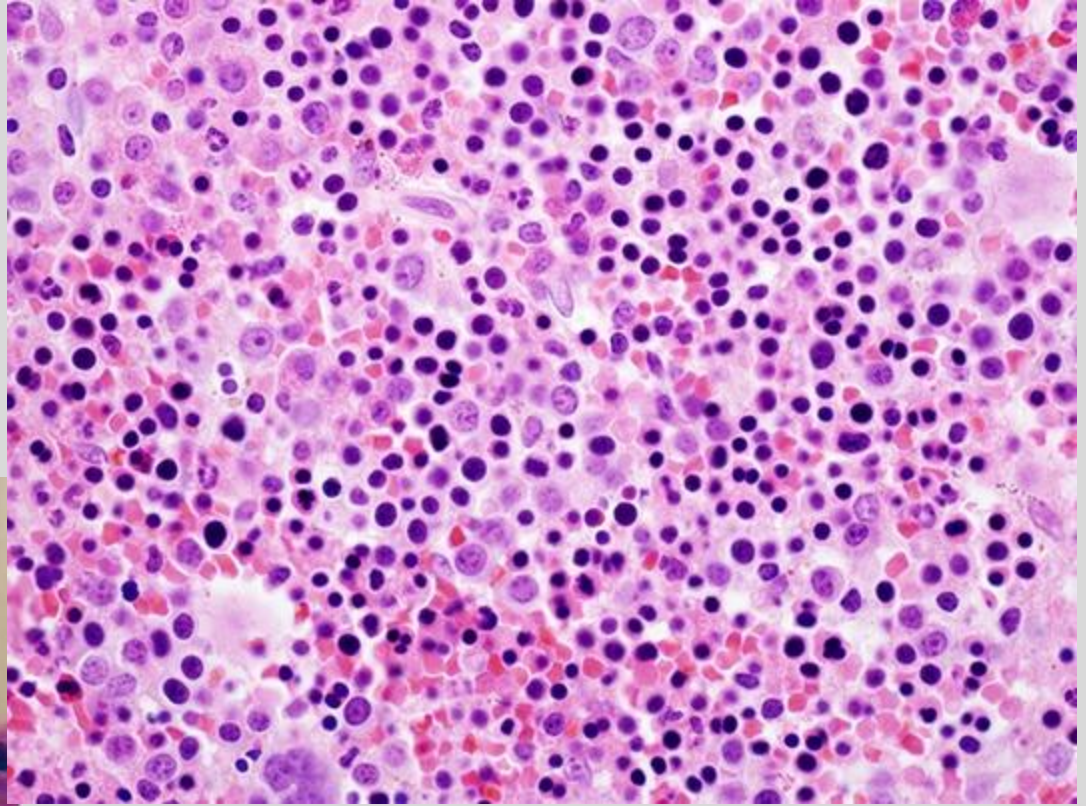




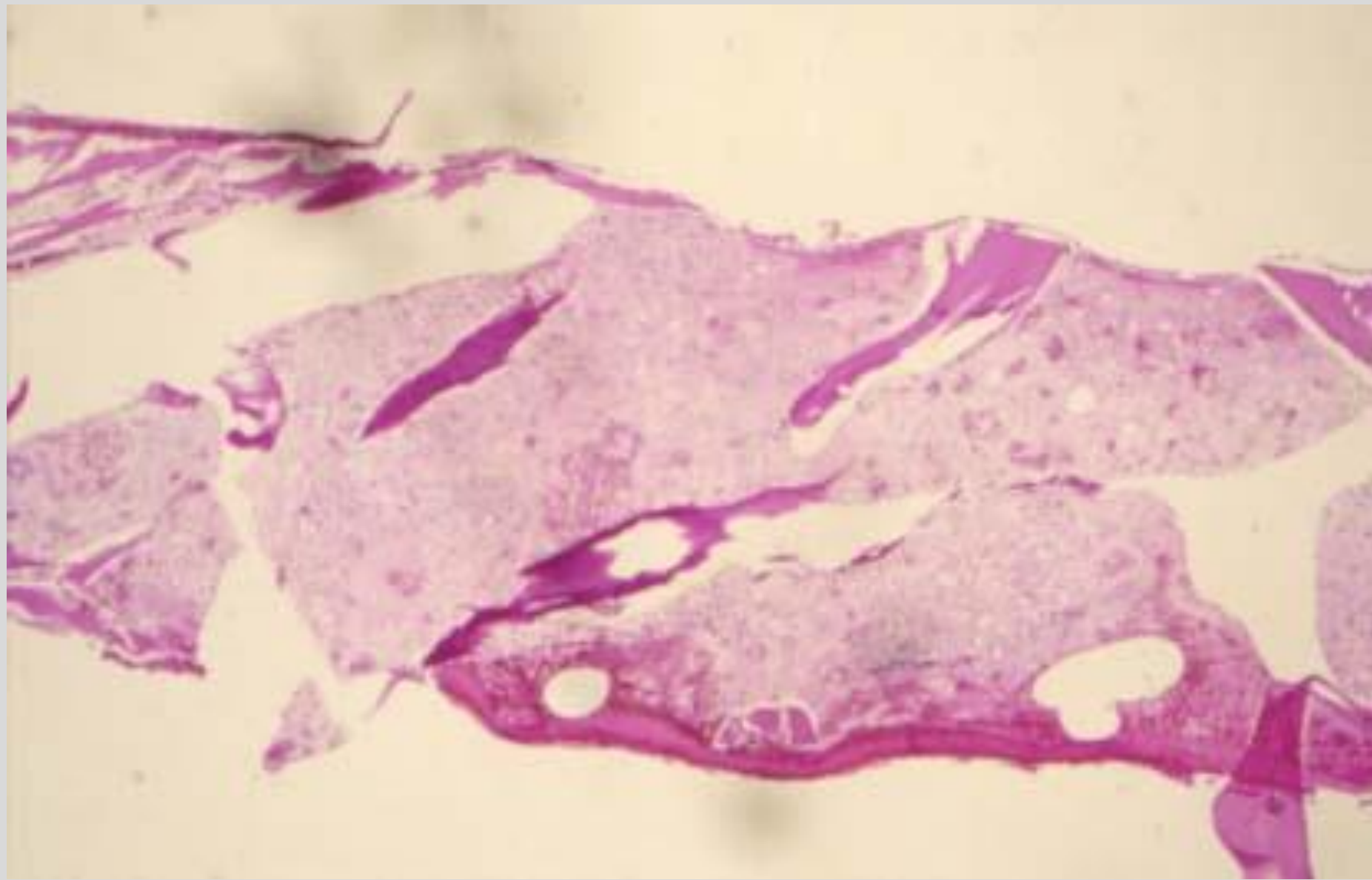
Questions

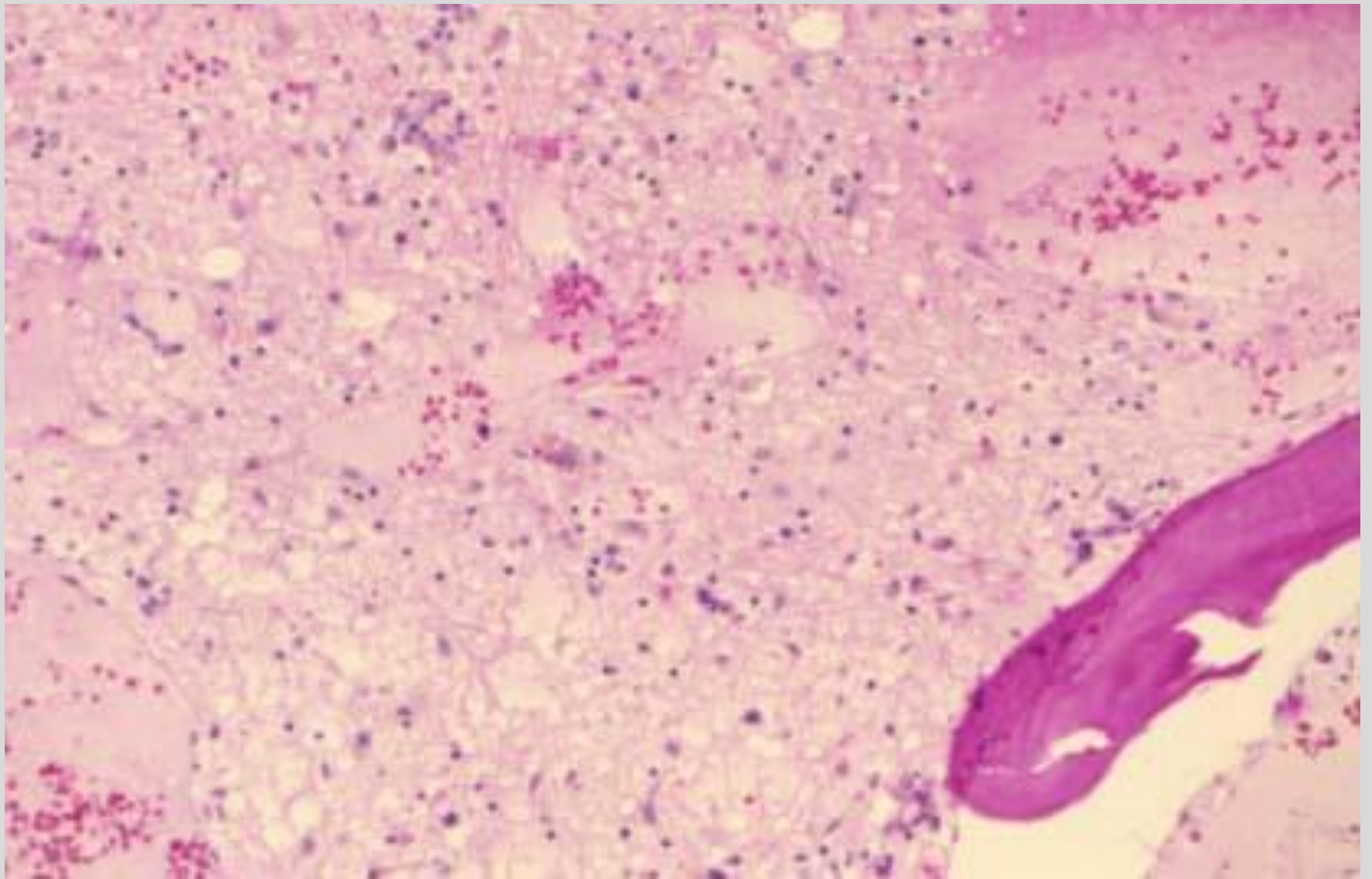
- What is a lymphocytosis?
- Why might this young girl present with bone pain?
- What are the different types of lymphocytes?

ERYTHROID HYPERPLASIA



Why would this occur?





Clinical Case

- A 3-year-old boy presents with a recent history of hives and wheezing after eating chocolate mousse. At 18 months of age, contact urticaria developed after he put his hand into a cake mix containing egg.
- He also has a history of chronic rhinitis, nocturnal cough, severe eczema, and a previous hospitalization for recurrent wheezing.

Clinical Case

- Which of the cell lines might you find elevated in this boy?
 - a) Eosinophils
 - b) Neutrophils
 - c) Megakaryocytes
 - d) Erythrocytes
- Other cell lines associated with allergy
 - Mast cell
 - Basophils
 - Plasma cells (IgE)

Normal CBC values

- RBC count:
 - M: 4.5 - 5.9 million/mm³
 - F: 4.0 - 5.2 million/mm³
- Hematocrit:
 - M: 41 - 53%
 - F: 36 - 46%
- Hemoglobin:
 - M: 13.5 - 17.5 g/dL
 - F: 12.0 - 16.0 g/dL
- Mean Corpuscular Volume (MCV)
 - 80 - 100 μm^3
- Mean Corpuscular Hemoglobin (MCH)
 - 26.0 - 34.0 pg/cell
- Mean Corpuscular Hemoglobin Concentration (MCHC)
 - 31.0 - 37.0 g/dL
- Reticulocyte count:
 - 0.5 - 2.5%
- Platelet count:
 - 150,000 - 350,000/mm³
- Leukocyte count with differential
 - Leukocyte count:
 - 4,500 - 11,000/mm³
 - Segmented Neutrophils
 - 40 - 70%
 - Bands
 - 0 - 10%
 - Eosinophils
 - 0 - 8%
 - Basophils
 - 0 - 3%
 - Lymphocytes
 - 22 - 44%
 - Monocytes
 - 4 - 11%

Definitions

- Anemia:
 - any condition where the number of erythrocytes (RBCs), Hemoglobin, or Hematocrit are decreased
- Polycythemia:
 - increase in total number of RBCs in peripheral blood
- Thrombocytopenia
 - abnormally low platelets in peripheral blood
- Thrombocytosis
 - increase in the number of platelets in the peripheral blood
- Monocytosis
 - increased number of monocytes in peripheral blood
- Eosinophilia
 - increased number of eosinophils in peripheral blood
- Leukopenia
 - total number of leukocytes in peripheral blood less than normal
- Leukocytosis
 - total number of leukocytes in peripheral blood greater than normal
- Neutropenia
 - abnormally small number of neutrophils in the peripheral blood
- Neutrophilia
 - an increased number of neutrophils in the peripheral blood
- Lymphopenia
 - a reduction, relative or absolute, in number of lymphocytes in peripheral blood
- Lymphocytosis
 - an increase, actual or relative, number of lymphocytes in peripheral blood

Definitions

- Hemoglobin (Hgb):
 - the red respiratory protein of RBCs, carries O₂ from lungs to tissue
- Hematocrit (Hct):
 - percentage of the volume of a blood sample occupied by cells
- Mean Corpuscular Volume
 - the average volume of RBCs
- Mean Corpuscular Hemoglobin
 - the hemoglobin content of the average RBC
- Mean Corpuscular Hemoglobin Concentration
 - the average hemoglobin concentration in a given volume of packed red blood cells
- Neutrophil = Polymorphonuclear cell (PMN)

Summary Slide

- Hematopoiesis – Fetus: yolk sac followed by hepatic phase; Infant all bone marrow; adult bone marrow mostly flat bones and proximal femur
- Red bone marrow – vascular, mainly newborns
- Yellow bone marrow – adipose tissue, adults
- Cellularity – normocellular, hypocellular, hypercellular, acellular, 100 – age +/- 10%
- Monophyletic theory of hematopoiesis - all blood cells arise from common stem cell, the pluripotential stem cell (CD34+), which gives rise to all other progenitor stem cells
- Bone marrow inability to keep up with demands results in extramedullary hematopoiesis in liver, spleen, lymph nodes

Summary Slide

- CFU – GEMM differentiates into CFU-GM, CFU-E, CFU-Meg, CFU-Eo, CFU-B; separate CFU-L (lymphocytes)
- Growth factors: CSFs (colony stimulating factor), IL (interleukin), erythropoietin, and thrombopoietin
- Review hierarchy of hematopoiesis for neutrophil, eosinophil, basophil, monocyte, red blood cell (erythrocyte), platelet (thrombocyte)
- Morphology and order of precursor cells – myeloblast>> promyelocyte>>myelocyte>>metamyelocyte>>band/stab>> PMN
- **Cannot differentiate eosinophil or basophil from neutrophil until late myelocyte/early metamyelocyte stage when specific granules appear**
- Morphology and order – proerythroblast>>basophilic erythroblast>>polychromatic erythroblast>> orthochromatic erythroblast>>reticulocyte>> erythrocyte (mature red blood cell)
- Thrombopoiesis: megakaryoblast>>Pro-megakaryocyte>> Megakaryocyte>>platelets

References

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